Sequences

1 For each of the following sequences, find the indicated term:

- **a** Third term in the sequence: $2, -4, 6, -8, 16, \dots$
- **b** Fourth term in the sequence: $3, 3.5, 4, 4.5, 5, 5.5, \dots$
- **c** Fifth term in the sequence: $5, 4, 3, 2, 1, 0, -1, \dots$

2 If T_n describes the nth term of the following sequences , find the indicated term:

- **a** T_3 in the sequence: $4, -5, 6, -7, 8, \dots$
- **b** T_4 in the sequence: $200, 20, 2, 0.2, 0.02, \dots$
- **c** T_5 in the sequence: $1, 2.5, 4, 5.5, 7, 8.5, \dots$
- **d** $T_3 + T_5$ in the sequence: $6, -8, 9, -10, 11, \dots$
- **e** $2T_2 - T_4$ in the sequence: $9, 12, 15, 18, 21, \dots$
- **f** $-4\left(T_3 + T_4\right)$ in the sequence: $1, 4, 5, 9, 14, 23, \dots$

3 In the sequence $5, -5, 7, -7, 9, -9, \dots$, find n if $T_n = 7$.

4 For each of the following sequences:

- **i** Describe the pattern in words.
- **ii** Find the next number in the sequence.

- **a** $-1, 1, 3, 5, 7, \dots$
- **b** $64, 32, 16, 8, 4, 2, \dots$
- **c** $2, -4, 6, -8, 10, -12, \dots$

Explicit rules

5 Determine whether the following are explicit relations:

- **a** $b_n = b_{n-1} b_{n-2}$
- **b** $d_n = n^2 + 3n + 8$
- **c** $a_n = 8a_{n-1} + a_1$
- **d** $c_n = 3n^2$

6 For each of the following explicit rules, state the first 5 terms of the sequences in order starting with $n = 1$:

- **a** $b_n = 5n - 2$
- **b** $s_n = n^2 + 6$
- **c** $t_n = 2n^2 + n - 3$

7 For the following sequencesn state the ex[illegible] rule that describes T_n in terms of n.

a A sequence starts with a first term of 1300 and each subsequent term increases by 2.5% of the previous term.

b A sequence starts with a first term of 44 and each term is 77 more than the previous term.